Patient-Aware Melanoma Classification under Severe Class Imbalance: Deep Visual Learning, Metadata Fusion and Reliability Analysis

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***Abstract*—Melanoma classification under severe class imbalance requires more than selecting the architecture with the highest accuracy. This study evaluates melanoma recognition on 33,126 SIIM-ISIC dermoscopic images, including only 584 malignant cases, using a fixed patient-aware validation design with zero patient overlap. Experiments progress from unweighted and class-weighted Logistic Regression baselines to ResNet18 (B0), ConvNeXt-Tiny (M1), and ConvNeXt-Tiny with leakage-safe clinical metadata fusion (M2). Class weighting increased Logistic Regression sensitivity but did not improve ranking. Under a matched deep-learning protocol, M1 improved ROC-AUC from 0.864 to 0.901 and Average Precision (AP) from 0.119 to 0.169 relative to B0, while sensitivity increased from 0.795 to 0.974. Metadata fusion did not improve ranking: M2 achieved ROC-AUC=0.897 and AP=0.165, but increased specificity from 0.628 to 0.736 and reduced false positives by 705 at threshold 0.5. Patient-level bootstrap, threshold analysis, subgroup evaluation, failure review and Grad-CAM further show that ranking performance alone does not characterise model reliability. The results emphasise patient-aware validation, imbalance-sensitive metrics and explicit error analysis over architecture benchmarking alone.**

***Index Terms*—Melanoma classification, class imbalance, ConvNeXt, metadata fusion, patient-aware validation, deep learning, Grad-CAM.**

I. Introduction

Automated melanoma classification is difficult because strong visual discrimination must be learned from data in which malignant examples are rare. Deep convolutional networks can learn diagnostically useful representations directly from skin-lesion images [1], but performance estimates remain sensitive to dataset construction and evaluation design. The SIIM-ISIC 2020 collection contains 33,126 dermoscopic images from 2,056 patients, of which only 584 (1.76%) are melanoma [2]. A classifier predicting benign for nearly every image can therefore obtain high accuracy while failing at the task of interest.

A second challenge is patient dependence. SIIM-ISIC is explicitly patient-centric and may contain several lesions from one patient [2]. Random image-level splitting can consequently mix patient-specific information across training and validation. This study uses a fixed patient-aware grouped split: 26,499 images for training and 6,627 for validation, with zero patient overlap. The training partition contains 467 melanomas and 26,032 benign images, an imbalance of approximately 1:55.7.

The reduced model set is intentional. Rather than increasing architectural diversity, the study prioritises comparisons in which the experimental change can be identified more clearly. ResNet18 and ConvNeXt-Tiny are both convolutional image encoders, while M2 preserves the ConvNeXt image path and adds only a small metadata branch. B0-to-M1 therefore asks whether replacing a conventional residual CNN with a modernised convolutional backbone changes performance under matched training and evaluation settings; M1-to-M2 asks whether clinical context changes an otherwise comparable ConvNeXt-based system. Adding unrelated architectures with different losses, sampling strategies or augmentation policies would return the study toward a leaderboard in which several design variables change simultaneously.

The study separates three questions. **RQ1:** how does class weighting affect minority detection in a fixed historical Logistic Regression (LR) baseline? **RQ2:** under a matched patient-aware protocol, how does ConvNeXt-Tiny compare with ResNet18? **RQ3:** does leakage-safe age, sex and anatomical-site metadata provide complementary information to ConvNeXt image features, and what does post-hoc analysis reveal about reliability? Contributions are: (i) patient-aware validation plus exact-content integrity auditing; (ii) a controlled B0-to-M1 architecture comparison and M1-to-M2 metadata ablation; and (iii) threshold, disagreement, paired patient-bootstrap, subgroup, failure and Grad-CAM analyses. This emphasis is important because strong internal discrimination alone does not establish clinical reliability or external validity [10].

II. Related Work

End-to-end CNNs established the viability of learned skin-lesion representations [1], while the SIIM-ISIC 2020 dataset added patient identifiers and clinical context suitable for patient-aware modelling [2]. Class weighting can counter rare-class under-emphasis, but improving one class can trade performance against another rather than improve every metric [3]. Under severe imbalance, precision-recall analysis is particularly useful because apparently strong accuracy or ROC behaviour can coexist with poor positive-class precision; this motivates reporting Average Precision (AP) alongside ROC-AUC [4].

Imbalance handling is deliberately uniform across the deep models. Oversampling and focal loss are valid alternatives, but assigning a different method to each architecture would make architecture and imbalance treatment inseparable. B0, M1 and M2 therefore share weighted BCE using the same training-derived positive weight. This converts class skew into an optimisation-cost adjustment while preserving the natural validation prevalence, so B0-to-M1 primarily changes the image encoder and M1-to-M2 primarily changes the information available to the classifier.

ResNet provides a well-established convolutional reference through residual mappings [5]. ConvNeXt is particularly suitable as the comparison architecture because it was developed by progressively modernising the ResNet design space rather than abandoning convolution [6]. Its design incorporates changes such as depthwise convolution, larger spatial kernels, inverted bottlenecks and Layer Normalisation while retaining a hierarchical convolutional representation. The comparison is therefore narrower than a CNN-versus-Transformer benchmark: both systems learn spatial feature hierarchies, but they represent different generations of convolutional design. B0 contains 11.18 M trainable parameters and M1 27.82 M, so capacity and several internal mechanisms change together; the experiment consequently supports an architecture-family comparison, not attribution to one ConvNeXt component. Resolution, training data, augmentation, loss, optimiser, class weighting, checkpoint criterion and primary threshold are held fixed.

Clinical context offers information not contained in lesion appearance alone. Gessert *et al.* fused patient metadata with deep image features for skin-lesion classification [7], and large ISIC challenge evaluations have also incorporated auxiliary clinical information [8]. Age, sex and anatomical site are therefore plausible complementary variables because they provide patient or lesion context that is not directly encoded as pixels. The present M1-to-M2 ablation nevertheless treats their value as empirical rather than guaranteed: metadata may change calibration or operating behaviour without improving case ranking. Late fusion is used because it preserves the pretrained image representation while allowing the comparatively low-dimensional clinical variables to be transformed separately before combination.

Patient-aware evaluation is equally important to this comparison. When multiple lesions arise from one patient, random image-level partitioning can make training and validation observations less independent than their image counts imply. Grouping by patient therefore addresses a dataset-structure problem rather than an architecture choice and is kept fixed for every experiment. Reliability is then examined with Grad-CAM, which provides coarse spatial attribution for a target logit [9]. Such attribution is descriptive rather than causal; faulty AI support can mislead clinicians despite strong headline performance [10].

III. Methodology

A. Dataset, Integrity and Validation

For image x_i, each model produces logit z_i, probability p_i and binary prediction

$$p_i = \sigma(z_i), \qquad \hat{y}_i = \mathbb{1}[p_i \geq \tau], \tag{1}$$

with common primary threshold $\tau = 0.5$. Five-fold `StratifiedGroupKFold` with seed 42 groups on patient identity; Fold 0 is permanently held as validation and folds 1–4 form training. This yields 26,499 training images (26,032 benign, 467 melanoma) and 6,627 validation images (6,510 benign, 117 melanoma), with zero patient overlap.

An exact-content audit SHA-256 hashed all 33,126 JPEGs. It found 433 duplicate hash groups (866 records), all within the same patient, split and target; no exact-content train-validation duplicate or conflicting-target group was found. This excludes byte-identical leakage only, not perceptual near-duplicates.

B. Baselines and Imbalance Handling

H0/H1 retain the historical LR representation: the 512×512 JPEG source images are converted to grayscale, resized to 64×64, scaled to $[0, 1]$, flattened to 4096 features, and trained on a stratified 5,000-image subset drawn only from the training partition. LR estimates

$$p_i = \sigma(\mathbf{w}^\top \mathbf{x}_i + b). \tag{2}$$

H0 is unweighted; H1 uses balanced class weights. B0/M1/M2 use the complete deep-learning training partition. The positive weight is training-derived,

$$w_+ = \frac{N_-}{N_+} = \frac{26032}{467} = 55.743, \tag{3}$$

and weighted BCE minimises

$$\mathcal{L}_{\mathrm{WBCE}} = -\frac{1}{N} \sum_i \left[w_+ y_i \log p_i + (1 - y_i) \log(1 - p_i) \right]. \tag{4}$$

Thus malignant errors receive greater optimisation weight without altering validation prevalence. The weight is determined from training labels rather than tuned against Fold 0.

C. Image Models and Metadata Fusion

The stored dataset is the 512×512 JPEG representation of SIIM-ISIC 2020, but B0, M1 and M2 resize RGB images to a common 224×224 network input. Holding resolution constant removes input size as an additional variable and reduces repeated fine-tuning cost. Training uses horizontal and vertical flips ($p = 0.5$), rotations up to 15°, and brightness/contrast jitter of 0.10. These moderate transforms reduce dependence on lesion orientation and introduce limited appearance variation without aggressively altering morphology. The policy is identical across all deep models; validation uses deterministic resizing and ImageNet normalisation only.

B0 uses ImageNet-pretrained ResNet18 [5]; residual blocks learn

$$\mathbf{y} = \mathbf{x} + \mathcal{F}(\mathbf{x}; \theta). \tag{5}$$

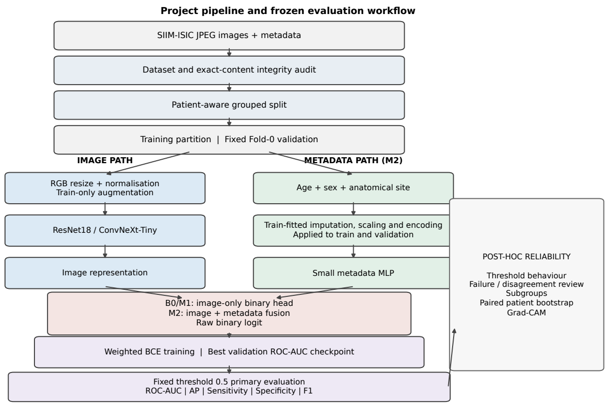

Fig. 1. Patient-aware pipeline and frozen evaluation workflow. Augmentation is training-only; metadata preprocessing is fitted on training data and applied without refitting to both partitions.

M1 replaces the backbone with ImageNet-pretrained ConvNeXt-Tiny [6]. Both terminate in a single raw-logit binary head.

M2 adds only age, sex and anatomical site; diagnosis fields, labels, image names and patient identifiers are excluded. Age uses train-fitted median imputation and standardisation; missing categorical values map to `unknown` before one-hot encoding. The resulting 11-D metadata vector is mapped to 32 dimensions and concatenated with the 768-D ConvNeXt representation:

$$\begin{aligned} \mathbf{v}_i &= h_\theta(x_i), \\ \mathbf{u}_i &= \mathrm{Dropout}(\mathrm{GELU}(\mathbf{W}_m m_i + \mathbf{b}_m)), \\ z_i &= \mathbf{w}^\top[\mathbf{v}_i; \mathbf{u}_i] + b, \quad p_i = \sigma(z_i). \end{aligned} \tag{6}$$

ResNet18 contains 11,177,025 trainable parameters, M1 27,820,897 and M2 27,821,313. M2 therefore adds only 416 parameters over M1: its purpose is to change the information source, not substantially increase capacity. M1 and M2 are independently initialised from ImageNet, so per-sample probability differences remain descriptive system disagreements rather than causal metadata effects.

D. Training and Evaluation

B0/M1/M2 use a deliberately shared configuration rather than model-specific hyperparameter optimisation. Batch size 32 was selected as a common memory-feasible setting for the available GPU and keeps the minibatch policy fixed across backbones. AdamW is used with learning rate 10^{-4} and weight decay 10^{-4}. The learning rate is intentionally conservative for full fine-tuning of ImageNet-pretrained representations, while the common weight decay supplies the same regularisation policy without coupling the comparison to different optimiser settings. No scheduler is used because architecture-specific schedules would add another optimisation variable; the aim is not to maximise each network independently but to preserve a comparable training policy. Automatic mixed precision reduces memory and arithmetic cost without changing the defined objective. All three use weighted BCE, a maximum of 10 epochs, patience 3 and validation ROC-AUC checkpoint selection with seed 42.

TABLE I
FIXED FOLD-0 VALIDATION RESULTS AT $\tau = 0.5$. H0/H1: LR; B0: RESNET18; M1: CONVNEXT-TINY; M2: CONVNEXT-TINY + METADATA.

Model	AUC	AP	Sens.	Spec.	F1
H0	0.657	0.039	0.009	0.998	0.016
H1	0.624	0.036	0.145	0.951	0.076
B0	0.864	0.119	0.795	0.768	0.108
M1	**0.901**	**0.169**	**0.974**	0.628	0.086
M2	0.897	0.165	0.897	**0.736**	**0.108**

The maximum epoch budget and stopping rule are also shared. B0 and M1 stopped after seven epochs and retained epoch 4; M2 completed ten epochs and retained epoch 8. Effective duration therefore follows the common checkpointing policy rather than a manually longer M2 schedule. No grid, random or Bayesian search was performed, so 224×224 resolution, batch size 32, learning rate, weight decay, patience, metadata width 32 and dropout 0.20 are design settings, not claimed optima. A stronger tuning design would require patient-grouped inner validation or nested cross-validation.

The primary decision threshold is fixed at $\tau = 0.5$ for every model. The later $\tau \in \{0.1, \ldots, 0.9\}$ sweep is descriptive rather than tuning: selecting a favourable threshold on Fold 0 and then reporting performance on the same fold would mix operating-point selection with evaluation. Metrics are ROC-AUC, AP, accuracy, balanced accuracy, precision, sensitivity, specificity, F1 and confusion counts. Reliability analysis includes threshold sweeps, M1/M2 disagreement, sex/age/site subgroups, 1,000 paired patient-level bootstrap replicates [11], qualitative failures and M2 Grad-CAM [9]. Fold 0 remains validation throughout; no independent test cohort is claimed.

IV. EXPERIMENTS AND RESULTS

A. Training Behaviour and RQ1: Class Weighting

Training curves are reported in Appendix Fig. 4. B0 and M1 attain their highest validation ROC-AUC at epoch 4 and subsequently fail to exceed that checkpoint before stopping at epoch 7. M2 improves later: its selected ROC-AUC maximum occurs at epoch 8, while minimum validation loss and highest AP occur at epoch 9. Epoch 8 is nevertheless restored because ROC-AUC was the predeclared checkpoint metric. This illustrates why one checkpoint cannot simultaneously optimise every metric.

H0 shows why accuracy is misleading: it achieved 98.1% accuracy but detected only 1/117 melanomas (sensitivity 0.009, specificity 0.998). H1 detected 17/117, increasing sensitivity to 0.145 while specificity fell to 0.951. Ranking did not improve: ROC-AUC changed 0.657→0.624 and AP 0.039→0.036. Weighting changes the relative optimisation cost of malignant errors and therefore shifts the fitted decision behaviour, but both LR variants still operate on the same flattened 64×64 grayscale pixels. **RQ1 is therefore supported only in minority-detection terms**: weighting reduces majority-class dominance without adding the visual information needed for stronger ranking.

B. RQ2: ResNet18 versus ConvNeXt-Tiny

The H1-to-B0 jump is a historical system-level progression because representation, training-set size, pretraining and augmentation all change. B0-to-M1 is the controlled architecture comparison. M1 increases ROC-AUC 0.864→0.901 (+0.037), AP 0.119→0.169 (+0.050), and sensitivity 0.795→0.974. It detects 114/117 melanomas compared with 93/117 for B0, reducing false negatives from 24 to 3. However, false positives rise from 1,513 to 2,424 and specificity falls 0.768→0.628.

ROC-AUC and AP assess ordering across continuous scores, whereas sensitivity and specificity at $\tau = 0.5$ describe one operating point. M1 therefore provides stronger ranking while its default threshold produces a larger false-positive burden. **RQ2 is supported for ranking and sensitivity, not universal operating-point performance.** The improvement also carries computational cost: B0 has 11.18 M parameters and trains for 593.76 s over seven epochs, versus 27.82 M and 767.50 s for M1; mean epoch time increases from 83.26 to 107.39 s. The result is an architecture-family improvement under matched settings, not an efficiency-normalised gain or causal evidence for one ConvNeXt mechanism.

C. RQ3: Metadata Fusion

M2 does not improve ranking: ROC-AUC changes 0.901→0.897 (-0.0034) and AP 0.169→0.165 (-0.0041). At $\tau = 0.5$, however, specificity rises 0.628→0.736 and false positives fall 2,424→1,719 (-705), while sensitivity falls 0.974→0.897 and false negatives rise 3→12 (+9). Of M1's false positives, 826 become M2 true negatives, while nine M1 true positives become M2 false negatives.

The near-zero ranking changes but much larger fixed-threshold changes suggest that the principal observed effect of the multimodal system is altered score placement around the common decision boundary rather than improved malignant-versus-benign ordering. Because the metadata branch adds only 416 parameters, this behaviour cannot be explained simply by a large increase in capacity. Nevertheless, M1 and M2 are independently trained, so individual transitions cannot be interpreted as causal contributions from age, sex or site. **RQ3 is therefore not supported as a ranking gain**: metadata instead changes the operating-point trade-off. Separate variable-level and repeated-seed ablations are required for stronger attribution.

M2 requires 1,080.50 s in total versus 767.50 s for M1, but this is mainly an epoch-count effect: M2 executes ten epochs versus seven. Mean epoch times are almost identical (106.43 versus 107.39 s), consistent with the small metadata branch adding negligible computational overhead per epoch.

D. Threshold Behaviour and Uncertainty

Fig. 2 shows M1/M2 behaviour from $\tau = 0.1$ to 0.9. At every evaluated threshold M1 is more sensitive and M2 more specific. At $\tau = 0.7$, M1 sensitivity/specificity are 0.855/0.771, versus 0.803/0.851 for M2. Increasing τ progressively suppresses positive predictions: specificity and precision rise while sensitivity falls. This confirms that the difference between M1 and M2 is not confined to one arbitrary cutoff; M2 remains

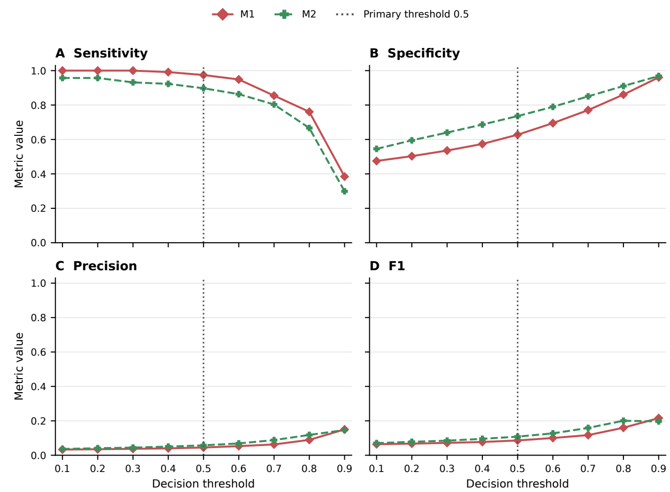

Fig. 2. M1/M2 sensitivity, specificity, precision and F1 across the post-hoc threshold grid. The dotted line marks $\tau = 0.5$; no threshold is selected as optimal.

the more conservative system throughout the inspected grid. Precision remains low across most thresholds despite high ROC-AUC, exposing the practical effect of 1.76% prevalence. In other words, good pairwise ranking does not remove the large number of benign images competing with each malignant case. No threshold is selected as optimal; 0.5 remains the common primary point.

Paired patient-level bootstrap gives M2-minus-M1 ROC-AUC difference -0.0034 with 95% percentile interval [-0.0204, 0.0130], and AP difference -0.0041 with [-0.0516, 0.0308]. Both cross zero, so the small ranking differences are not stable in direction across internal patient-level resamples. Sensitivity difference is -0.0769 with [-0.1395, -0.0261], showing a more persistent reduction in melanoma detection for M2 within this cohort. Patient rather than image is the resampling unit so that repeated lesions from one individual move together. These intervals quantify internal Fold-0 sampling variation only; they are not external validation, statistical equivalence, or proof of clinical superiority.

E. Failure, Subgroup and Attribution Analysis

Qualitative failure review found hair and marker-like artifacts among both correct and incorrect cases, providing no single artifact-driven explanation of error (Appendix Fig. 8). High-confidence errors include both visually subtle lesions and apparently conspicuous pigmented regions, while similar acquisition artifacts also appear among correct predictions. The failure montage is therefore used to generate hypotheses rather than assign a causal failure mechanism.

M2 specificity is higher across every displayed sex, age and site subgroup, while sensitivity is lower for both sexes and all age bands (Appendix Fig. 10). Several site groups contain only 1–3 melanomas, so apparent perfect sensitivity cannot support fairness claims; reporting positive support beside subgroup metrics is essential to avoid over-interpreting unstable percentages. Grad-CAM maps sometimes overlap visible lesions but can also extend into background, hair or marked regions (Appendix Fig. 9). Because maps target the raw melanoma logit with metadata supplied during the forward pass, they are image-branch attribution conditioned on metadata, not an explanation of the metadata branch or of causal reasoning. Together, the subgroup and attribution analyses support caution: aggregate discrimination can conceal both operating-point and case-level heterogeneity.

V. Limitations and Future Work

Fold 0 is a fixed patient-aware validation fold, not an independent test cohort. It was used for checkpoint selection and reused for exploratory threshold, subgroup, failure, bootstrap and Grad-CAM analyses; the reported metrics therefore describe internal validation rather than unbiased external generalisation. Full patient-grouped cross-validation and evaluation on an independent cohort are the strongest next steps [8], [10].

The controlled configuration trades model-specific optimisation for experimental interpretability. A shared resolution, batch size, learning rate, regularisation strength, loss and stopping rule makes B0/M1/M2 easier to compare, but no experiment establishes that 224×224, batch size 32, learning rate 10^{-4} or weight decay 10^{-4} are individually optimal. Likewise, M2's metadata embedding width 32 and dropout 0.20 are predeclared design choices, not tuned values. Future optimisation should therefore use patient-grouped inner validation or nested cross-validation rather than adapting settings to the reported Fold 0.

Metadata variables were introduced jointly, preventing variable-level attribution, and several subgroup analyses contain very few malignant cases. SHA-256 auditing detects byte-identical images only, while Grad-CAM remains coarse non-causal attribution. Future work should prioritise independent evaluation, nested tuning, repeated-seed and variable-level metadata ablations, perceptual duplicate checks and richer multimodal attribution rather than merely adding architectures.

VI. Conclusion

Patient-aware melanoma classification was reframed as controlled problem solving rather than architecture ranking. For RQ1, weighting increased LR sensitivity from 0.009 to 0.145 but did not improve ranking. For RQ2, ConvNeXt-Tiny improved ResNet18 ROC-AUC from 0.864 to 0.901 and AP from 0.119 to 0.169, while its $\tau = 0.5$ operating point sacrificed specificity for sensitivity. For RQ3, metadata fusion did not improve ranking but shifted behaviour toward higher specificity, reducing false positives by 705 while introducing nine additional false negatives. Bootstrap, threshold, subgroup, failure and Grad-CAM analyses show that aggregate ROC-AUC alone does not characterise reliability. The central result is therefore methodological: patient grouping, imbalance-sensitive metrics and explicit error analysis are as important as architecture choice when interpreting melanoma classifiers.

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Appendix A
Dataset Distribution and Integrity

The full dataset, training partition and Fold-0 validation partition retain similar melanoma prevalence despite severe imbalance. The SHA-256 audit found 433 exact duplicate pairs among 33,126 images; all remained within the same patient, split and target, with no exact-content train-validation leakage.

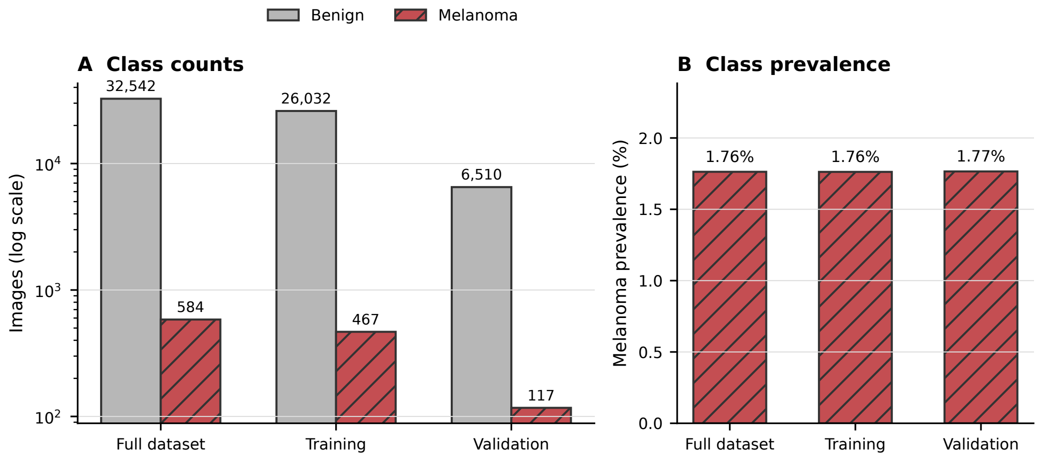

Fig. 3. Class distribution across the complete dataset, training partition and fixed Fold-0 validation partition. Melanoma prevalence remains approximately 1.76%.

APPENDIX B
EXPERIMENTAL CONFIGURATION

TABLE II

Deep-model protocol.

Model	Architecture	Pretraining	Image	Batch	Loss	w_+	Optimizer	LR	WD	Max ep.	Pat.	Run ep.	Best ep.	Params
B0	ResNet18	ImageNet V1	224	32	weighted BCE	55.743	AdamW	10^{-4}	10^{-4}	10	3	7	4	11,177,025
M1	ConvNeXt-Tiny	ImageNet V1	224	32	weighted BCE	55.743	AdamW	10^{-4}	10^{-4}	10	3	7	4	27,820,897
M2	ConvNeXt-Tiny+meta	ImageNet V1	224	32	weighted BCE	55.743	AdamW	10^{-4}	10^{-4}	10	3	10	8	27,821,313

TABLE III

Hyperparameter-selection strategy. No systematic grid, random or Bayesian search was performed.

Parameter	B0	M1	M2	Selection basis
Image size	224	224	224	predeclared shared
Batch size	32	32	32	predeclared shared
Optimizer / LR	AdamW / 10^{-4}	AdamW / 10^{-4}	AdamW / 10^{-4}	predeclared shared
Weight decay	10^{-4}	10^{-4}	10^{-4}	predeclared shared
Loss / w_+	WBCE / 55.743	WBCE / 55.743	WBCE / 55.743	shared / data-derived
Max epochs / patience	10 / 3	10 / 3	10 / 3	predeclared shared
Best checkpoint	4	4	8	max validation ROC-AUC
Primary threshold	0.5	0.5	0.5	predeclared shared
Metadata embedding / dropout	–	–	32 / 0.20	architecture-specific
Threshold grid	–	0.1–0.9	0.1–0.9	post-hoc analysis only

TABLE IV

M1-to-M2 metadata-fusion ablation at $\tau = 0.5$.

Metric	M1	M2	M2-M1
ROC-AUC	0.901	0.897	-0.003
Average Precision	0.169	0.165	-0.004
Accuracy	0.634	0.739	0.105
Balanced accuracy	0.801	0.817	0.016
Precision	0.045	0.058	0.013
Sensitivity	0.974	0.897	-0.077
Specificity	0.628	0.736	0.108
F1	0.086	0.108	0.022
FP / FN	2424 / 3	1719 / 12	-705 / +9

APPENDIX C
TRAINING AND CLASSIFICATION BEHAVIOUR

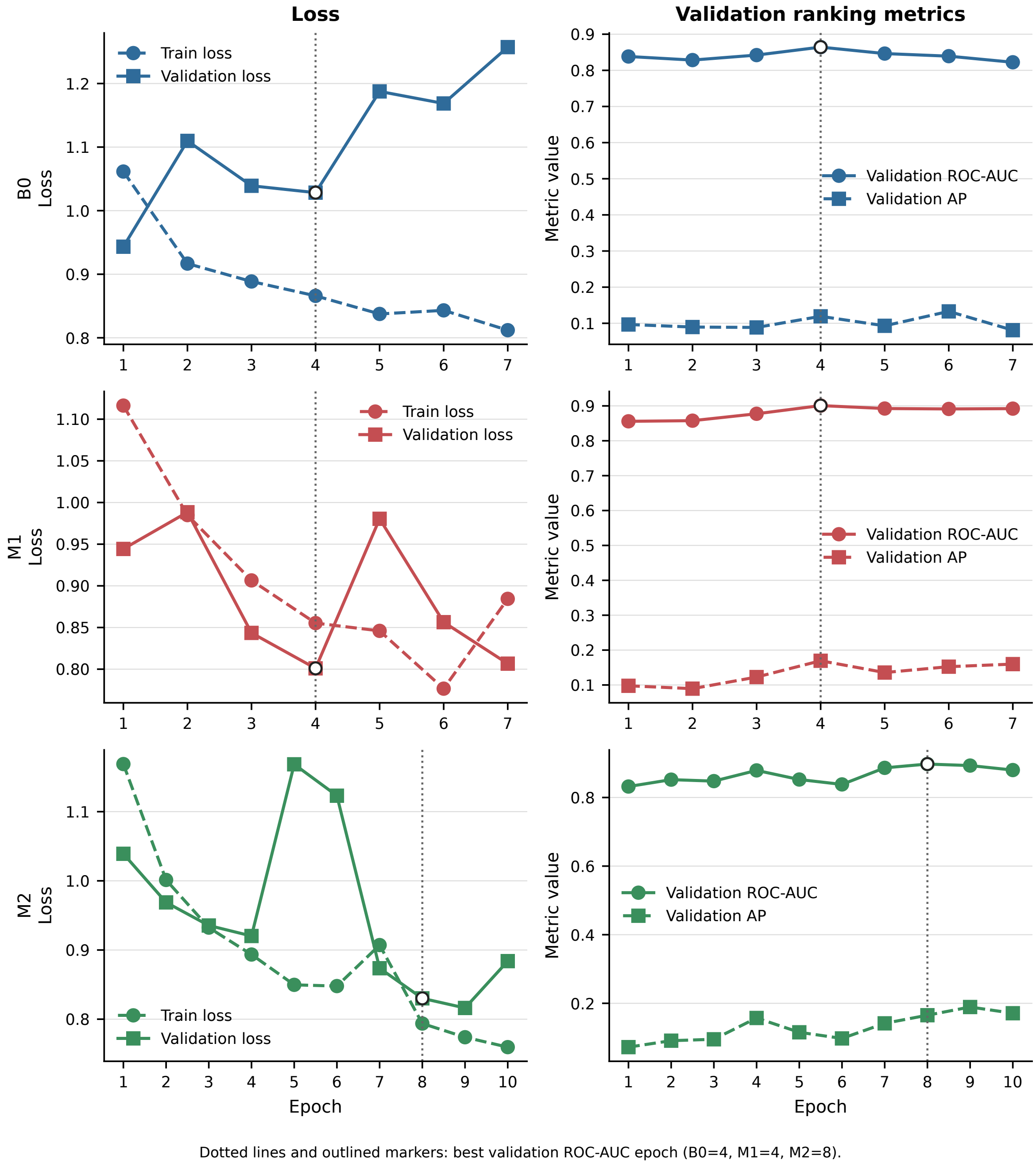

Fig. 4. B0/M1/M2 training loss, validation loss, ROC-AUC and AP. Dotted markers indicate the retained maximum-validation-ROC-AUC checkpoint.

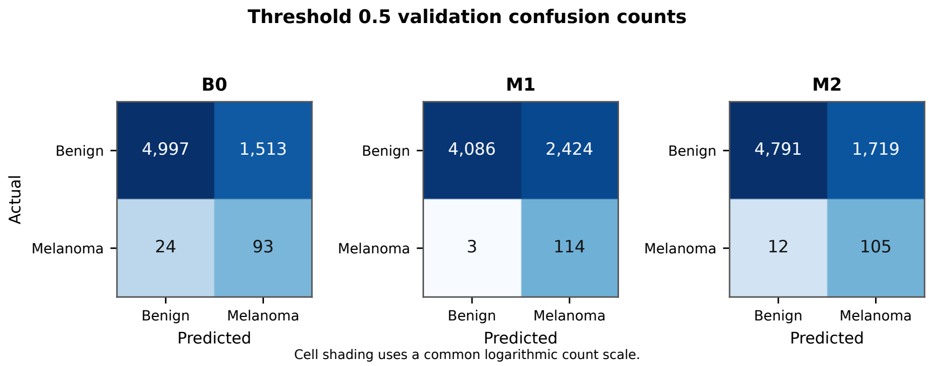

Fig. 5. Validation confusion matrices for B0, M1 and M2 at $\tau = 0.5$.

Appendix D
Ranking and Bootstrap Uncertainty

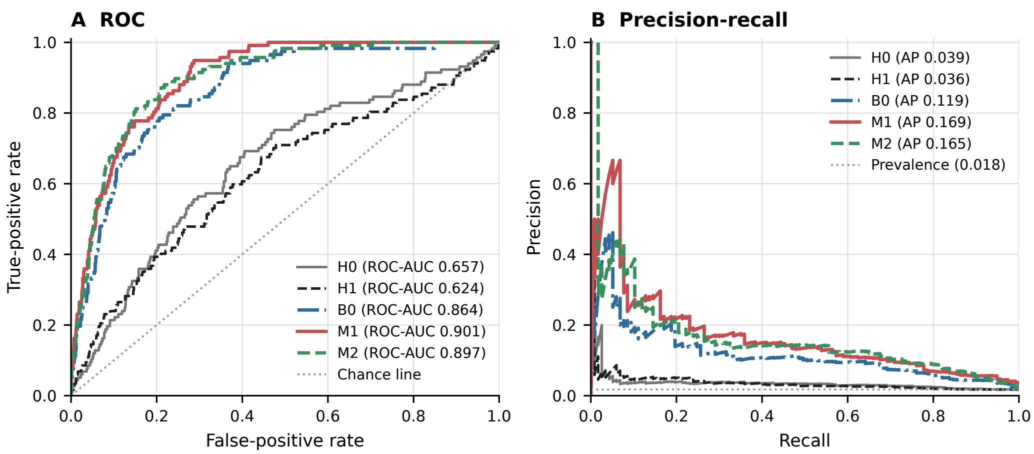

Fig. 6. ROC and precision-recall curves for H0, H1, B0, M1 and M2. Precision-recall legends report Average Precision (AP); the dashed baseline denotes melanoma prevalence.

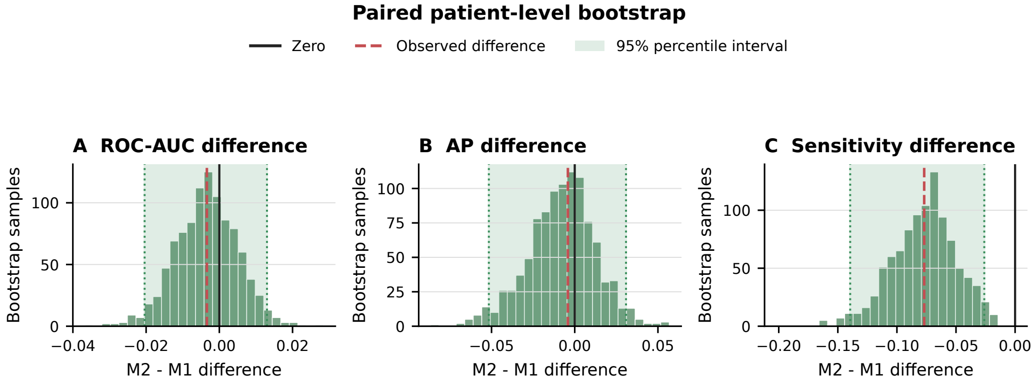

Fig. 7. Paired patient-level bootstrap distributions for M2-minus-M1 ROC-AUC, AP and sensitivity differences over 1,000 valid replicates. Zero denotes no difference.

TABLE V

Paired patient-level bootstrap summary.

Metric	Observed M2-M1	Median	2.5th pct.	97.5th pct.
ROC-AUC	-0.003	-0.003	-0.020	0.013
Average Precision	-0.004	-0.007	-0.052	0.031
Sensitivity	-0.077	-0.075	-0.140	-0.026

APPENDIX E
FAILURE ANALYSIS AND EXPLAINABILITY

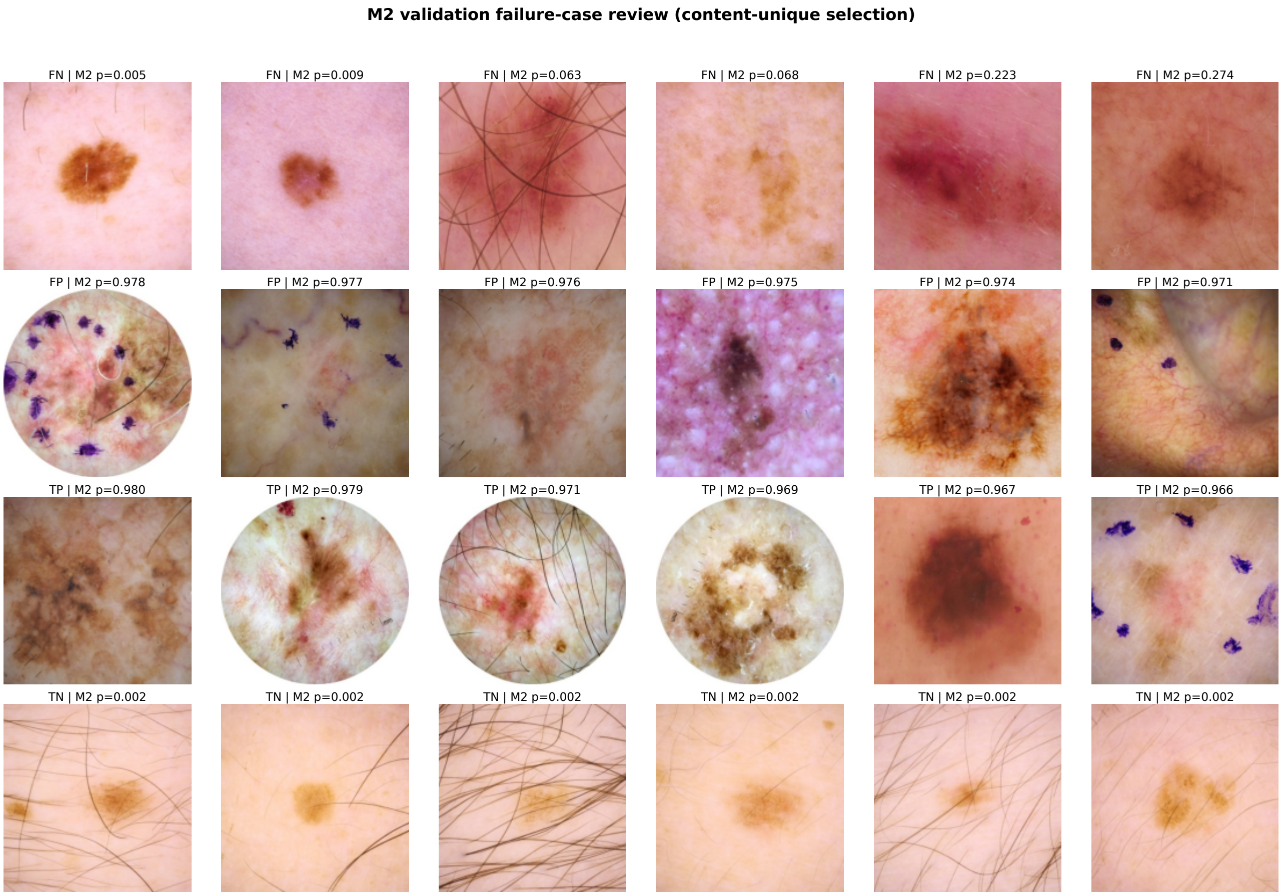

Fig. 8. Content-unique M2 false-negative, false-positive, true-positive and true-negative examples from the fixed validation fold. Displayed probabilities are frozen model outputs.

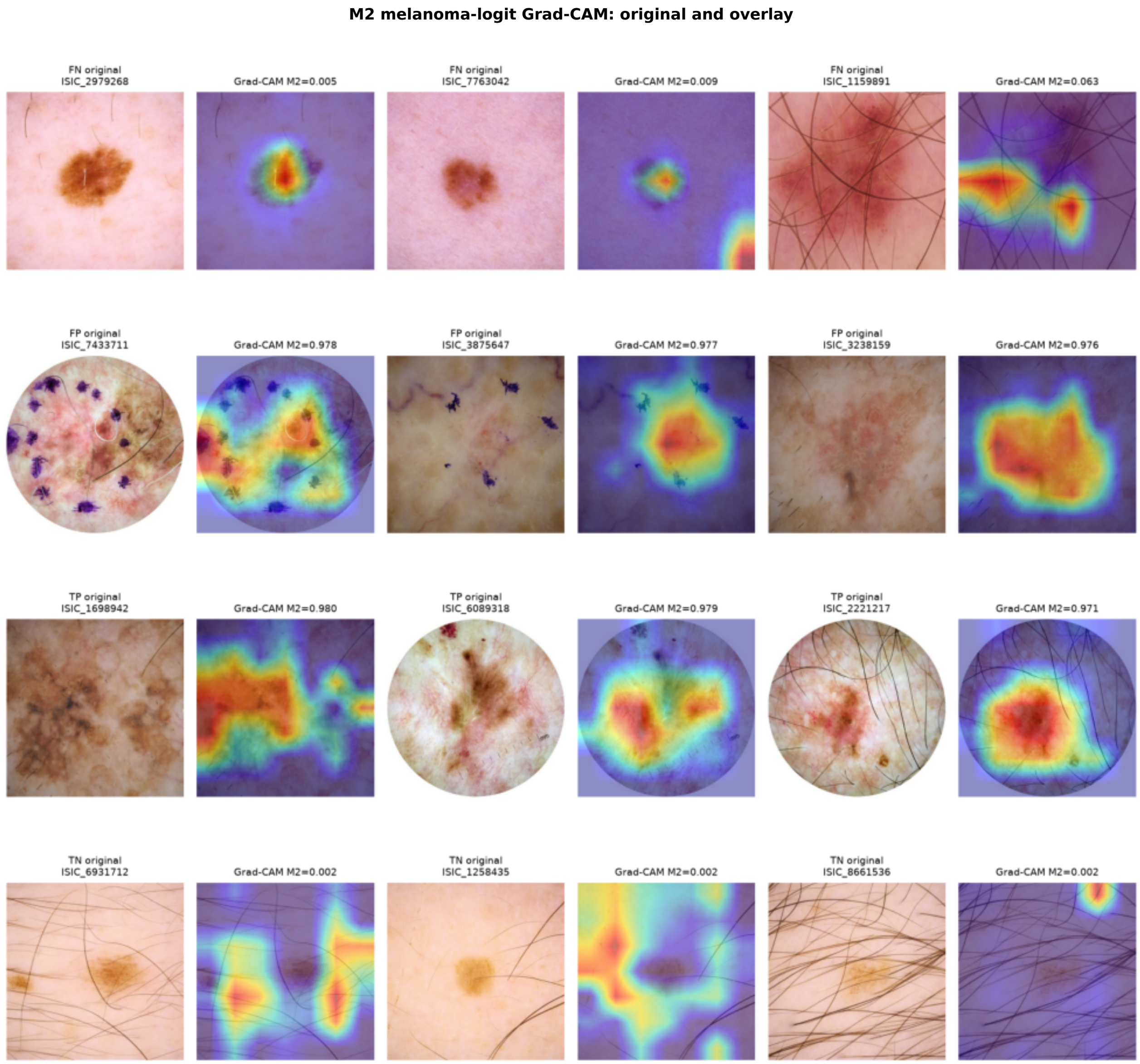

Fig. 9. M2 melanoma-logit Grad-CAM examples. Heatmaps represent image-branch spatial attribution conditioned on metadata; they do not explain the metadata branch.

APPENDIX F
SUBGROUP ANALYSIS

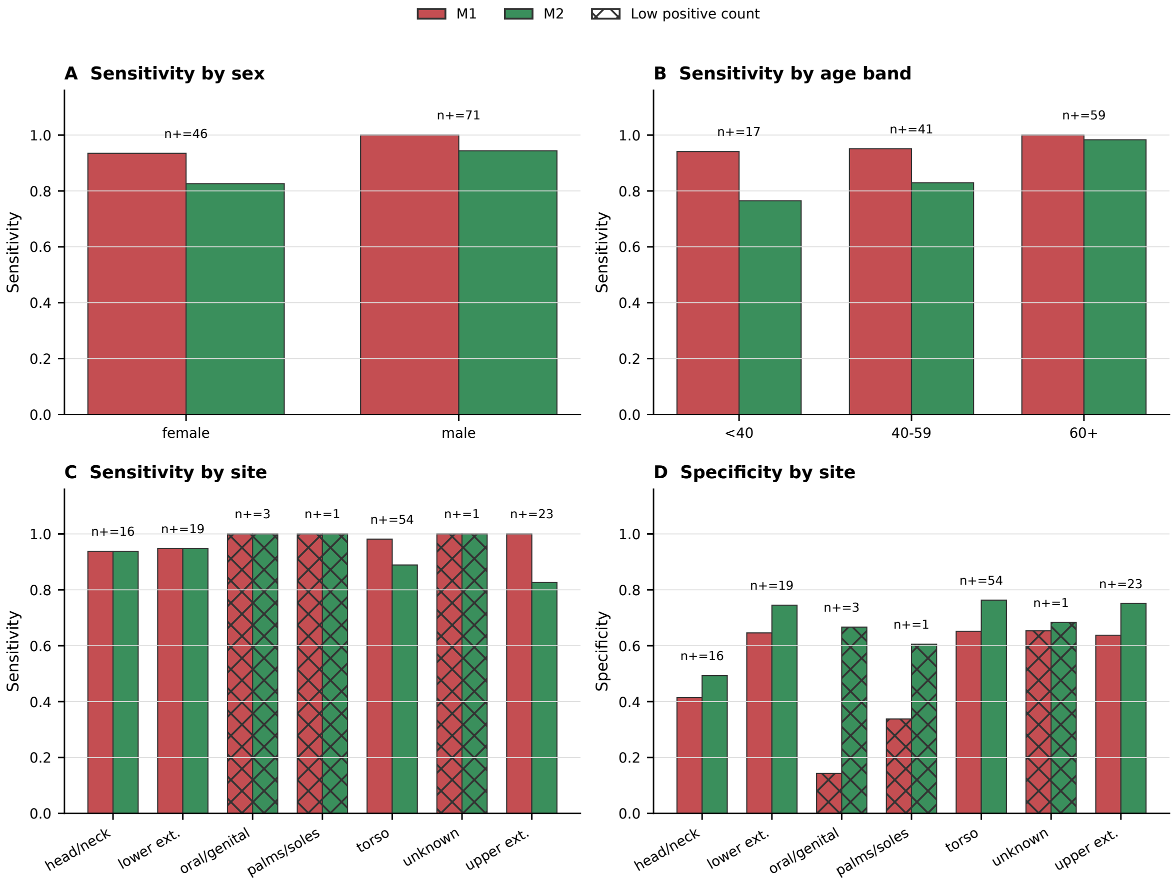

Fig. 10. M1/M2 sensitivity and specificity by sex, age band and anatomical site. Positive-case counts are displayed; hatching denotes low malignant support.

TABLE VI

Selected subgroup results; low-positive site categories are retained to expose uncertainty.

Model	Variable	Group	N	Positives	Sensitivity	Specificity
M1	sex	female	3332	46	0.935	0.641
M2	sex	female	3332	46	0.826	0.738
M1	sex	male	3295	71	1.000	0.614
M2	sex	male	3295	71	0.944	0.734
M1	age	< 40	1432	17	0.941	0.625
M2	age	< 40	1432	17	0.765	0.780
M1	age	40–59	2964	41	0.951	0.653
M2	age	40–59	2964	41	0.829	0.771
M1	age	60+	2231	59	1.000	0.595
M2	age	60+	2231	59	0.983	0.660
M1	site	oral/genital	24	3	1.000	0.143
M2	site	oral/genital	24	3	1.000	0.667
M1	site	palms/soles	72	1	1.000	0.338
M2	site	palms/soles	72	1	1.000	0.606
M1	site	unknown	102	1	1.000	0.653
M2	site	unknown	102	1	1.000	0.683